

# Research Methods for Transient Stability Analysis of Power Systems under Large Disturbances

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**Abstract:** Transient stability analysis is critical for maintaining the reliability and security of power systems. This paper provides a comprehensive review of research methods for transient stability analysis under large disturbances, detailing the modeling concepts and implementation approaches. The research methods for large disturbance transient stability analysis are categorized into five main types: simulation methods, direct methods, data-driven methods, analytical methods, and other methods. Within the analytical method category, several common analytical strategies are introduced, including the asymptotic expansion method, intrusive approximation method, and other analytical methods. The fundamental principles, characteristics, and recent research advancements of these methods are detailed, with particular attention to their performance in various aspects such as computational efficiency, accuracy, applicability to different system models, and stability region estimation. The advantages and disadvantages of each method are compared, offering insights to support further research into transient stability analysis for hybrid power grids under large disturbances.

**Keywords:** transient stability analysis; large disturbance; power systems; qualitative analysis methods



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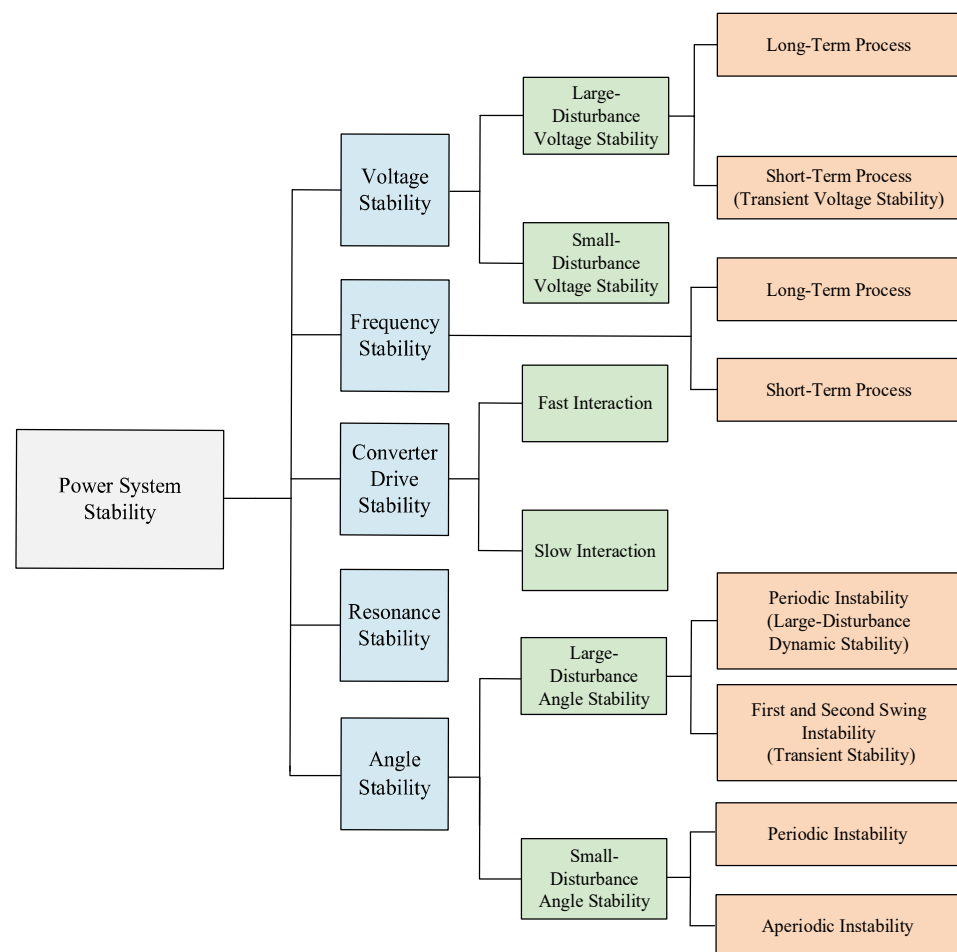
## 1. Introduction

With the widespread integration of large-scale high-capacity units, the rapid expansion of transmission networks, and the increasing demand for various types of electrical loads, the structural characteristics of the power grid have become more complex, and the interactions among components have intensified [1]. This complexity intertwines the dynamic responses of rotor angles and voltages following disturbances, interweaving stability modes and posing new challenges for power system stability analysis and operational control [2]. In practical operations, power systems inevitably experience various disturbances such as short-circuit faults, line disconnections, and the integration or disconnection of large-capacity loads. These disturbances frequently lead to widespread blackouts globally [3,4]. A review of past blackout incidents reveals common characteristics: an initial disturbance induces changes in the power grid topology, leading to power imbalances and flow shifts. These phenomena cause abnormalities in system parameters such as power angles, voltage, and frequency. Consequently, multiple electrical devices trigger fault alarms and intermittently disconnect, creating positive feedback loops and secondary shocks. This progression ultimately evolves into cascading failures [5], resulting in stability issues in the power system and causing significant impacts on the national economy and social governance.

Therefore, researching power system transient stability is crucial for preventing widespread blackouts and ensuring the safe and reliable operation of power grids. In fact, the need for stability research was formally recognized as early as the advent of industrial society. In 1868, James Clerk Maxwell, a pioneer in electromagnetism, studied the instability of amplitude output in centrifugal governor devices using differential equations of basic systems [6], thus pioneering the study of linear system stability. Similarly, in the

field of power systems, as early as the 1910s, electricity generated by hydropower plants in remote areas was transmitted over long distances to urban load centers [7]. Due to the significant costs of laying transmission lines, power suppliers aimed to deliver as much power as possible, often operating transmission lines in an overloaded state. Even minor disturbances could trigger power flow oscillations, causing the grid to lose stability [8]. Since then, power system stability has gradually come to the forefront, becoming a critical issue in power systems and continuing to be so to this day.

Particularly since the 21st century, rapid advancements in engineering and technology have led to a leap in the evolution of grid structures. The invention of flexible DC technology has enabled the integration of wind, photovoltaic, and thermal power, resulting in a grid structure characterized by both centralized and decentralized configurations. The commissioning of numerous High-Voltage Direct Current (HVDC) projects has created a “multi-infeed” AC-DC system [9]. With the large-scale integration of centralized wind farms and photovoltaic power plants [10,11], humanity has entered a new era of hybrid grids that feature multiple large-capacity DC transmission lines and large-scale wind and photovoltaic power [12,13]. This has profoundly transformed the dynamic behavior of the grid, making the security and stability characteristics and mechanisms of the grid increasingly complex, thus enriching the concept of power system stability. Consequently, power system stability has become increasingly diverse, and detailed classifications are illustrated in Figure 1 below.



**Figure 1.** Classification of power system stability.

Power system stability is defined as the ability of a power system to maintain stable operation after being subjected to disturbances. Based on the physical characteristics of instability, large-disturbance stability is further subdivided into large-disturbance angle

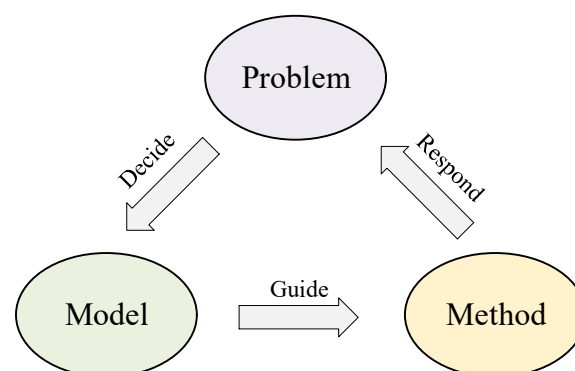
stability, large-disturbance voltage stability, and large-disturbance frequency stability. These categories focus respectively on the ability of generators to remain synchronized after large disturbances, the ability to keep bus voltages within acceptable limits, and the ability to maintain system frequency within acceptable bounds. Although in practical engineering these instability phenomena do not always occur in isolation—angle instability, voltage instability, and frequency instability often interrelate—differentiating large-disturbance stability into these distinct modes helps to address the main issues, simplify the problem scope, and facilitate the analysis of the primary causes of instability in various scenarios, thereby guiding engineering practice.

In general, the main task in studying power system stability is to solve and analyze initial value problems involving differential-algebraic equations with parameter  $u$ . These equations consist of state differential equations, network algebraic equations, output algebraic equations, and initial state conditions, as illustrated in Equation (1) below.

$$\begin{cases} \dot{X} = f(X, V, u) \\ YV = I(X, V, u) \\ y = h(X, V, u) \end{cases}, \quad (1)$$

where  $X$  represents the state vector describing the system's dynamic characteristics;  $Y$  denotes the node admittance matrix of the power network;  $V$  indicates the voltage vector at the network nodes;  $y$  represents the output variables of the grid operation;  $u$  encompasses the control parameters of the power system, including various grid operation parameters such as synchronous machine outputs, external faults, the integration status of wind and photovoltaic power plants, and deployable flexible equipment;  $f$  and  $I$  denote the dynamic equations of the network's dynamic components and the network's injected currents, respectively; and  $h$  represents the output function.

Mathematically, the essence of power system stability analysis is the quantitative characterization of how the parameter  $u$  affects the state variables  $X$ , output variables  $y$ , initial state conditions  $X_0$ , dynamic equation structure  $f$ , output function  $h$ , injected currents  $I$ , and admittance matrix  $Y$ . This is known as parameter stability analysis. Thus, in the context of Equation (1), power system stability analysis involves three research dimensions: problem, model, and method, as illustrated in Figure 2. These dimensions are closely interconnected and mutually influence each other.



**Figure 2.** Research dimensions in power system stability analysis.

The first dimension concerns the research questions, encompassing three aspects. Firstly, it involves investigating the fundamental characteristics of the state curve  $X(t)$  and the output curve  $y(t)$ . Depending on the type of electrical quantities of interest in the curve  $y(t)$ , the basic instability modes of the power grid can be categorized as voltage, power angle, frequency instability, or phase-locked loop (PLL) loss of lock [14,15]. Examining the short-term or long-term dynamic behaviors of the state curve and output curve, such as the coherence, multi-timescale, positivity, and convexity of the output variable  $y$ , aids in under-

standing the stabilization mechanisms of the power grid. Secondly, it involves exploring the fundamental structure of equations, including the inherent structure and mathematical properties of the vector field function  $f(X, V, u)$ , the output function  $h(X, V, u)$ , and the network equation  $YV = I(X, V, u)$ . This stems from the profound insights gained from practical experiences with power grids, such as the fact that synchronous compensators are more beneficial for voltage stability compared to static reactive compensation, and that photovoltaics, due to their constant power characteristics, cannot support voltage like synchronous machines. These well-established operational experiences lead to the belief that the equation structure underlying the dominant physical phenomena in power systems should not be chaotic. Reference [16] demonstrated that power systems can be approximated as hybrid monotone systems and revealed the inherent growth and decay mechanisms in system structure through the sign characteristics of the power grid equation Jacobian matrix at different time periods. Reference [17] clarified the all-positive and block-diagonal dominance properties of the voltage-reactive power Jacobian matrix in the network equation, explaining existing engineering understanding. Thirdly, it involves investigating the influence of the parameter  $u$  on state curves  $X(t)$  and  $y(t)$ , i.e., the stability mechanisms, including the analytical determination of the state space stability region  $S$ , the stability boundary  $\partial S$ , the stability margin, and the response curves  $X(t; u)$  and  $y(t; u)$ .

The second research dimension is the equation model. The mathematical models for stability analysis originate from engineering practice and are specific, generally unmodifiable, and encompass three aspects. Firstly, they involve specifying mathematical models for various control elements of the power grid, such as speed regulators and power system stabilizers (PSSs), under overexcitation limits. The main goal is to analyze how changes in the parameters of these control models affect the stability of the power grid [18]. Secondly, they involve specifying component models. This includes models of various generation, transmission, distribution, and consumption elements such as wind turbines, frequency-controlled air conditioners, electric vehicles, and controllable series compensators. The focus is on analyzing the impact of integrating these components on the stability of the power grid [19]. Thirdly, they involve specifying the operational modes of the power grid. This aspect explores how operational modes affect the base case power flow, i.e., how different operating conditions impact the initial value  $X_0$  in stability analysis, as exemplified by classical emergency control [20] and preventive control [21].

The third research dimension is the methodological aspect, which focuses on identifying specific analysis methods to address problems based on the given mathematical models of power systems. Over nearly a century of development, transient stability analysis in power systems has evolved into various approaches [22,23], as shown in Figure 3. These methods explore the nonlinear behavior of power systems following major disturbances from different angles, which is a key challenge in transient stability analysis. For instance, simulation methods use step-by-step integration to directly obtain numerical solutions and capture the system's nonlinear dynamics. Direct methods examine the relationship between kinetic and potential energy from an energy perspective under nonlinear conditions. Analytical methods aim to derive approximate analytical expressions  $X(t; u)$  and  $y(t; u)$  for nonlinear system responses and explore quantitative relationships between the parameter  $u$  and stability. These systematic research methods vary in their analytical depth, precision, and intuitiveness, as illustrated in Figure 4. When combined with long-term practical experience from power grid operations—such as the observation that units with higher inertia are less prone to transient stability issues and networks with strong electrical distance characteristics are more likely to maintain stability—these methods provide a comprehensive understanding of transient stability in power systems.

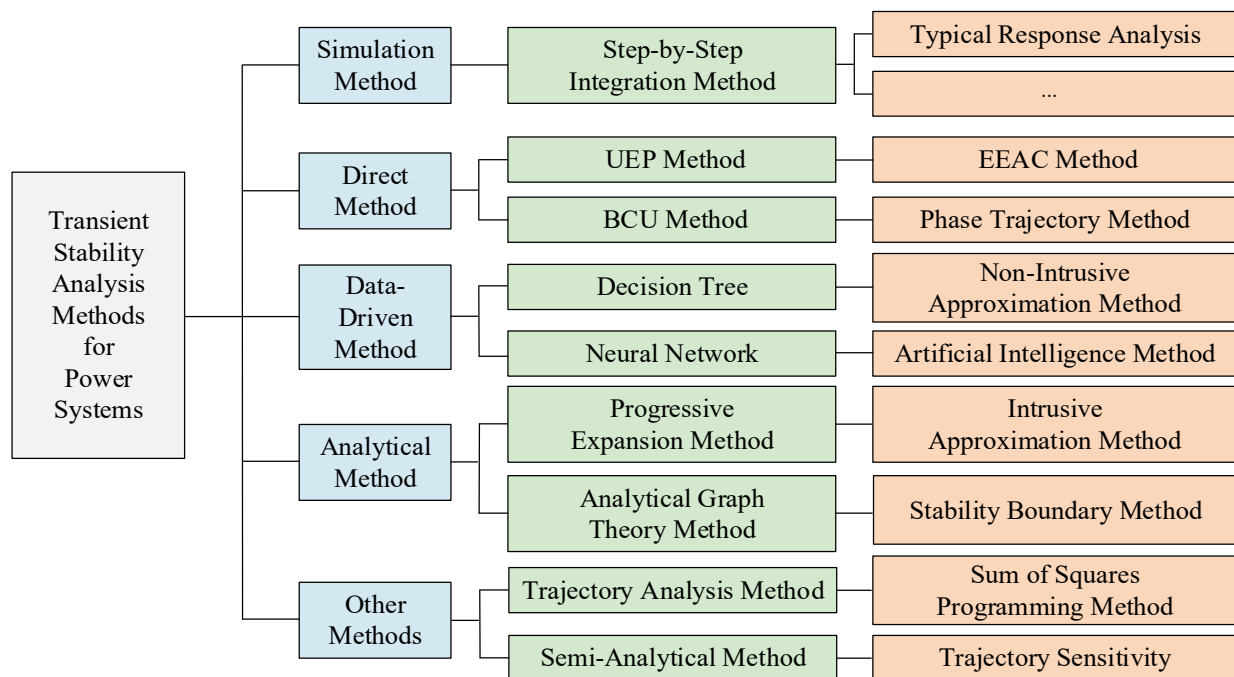


Figure 3. Classification of research methods for transient stability analysis in power systems.

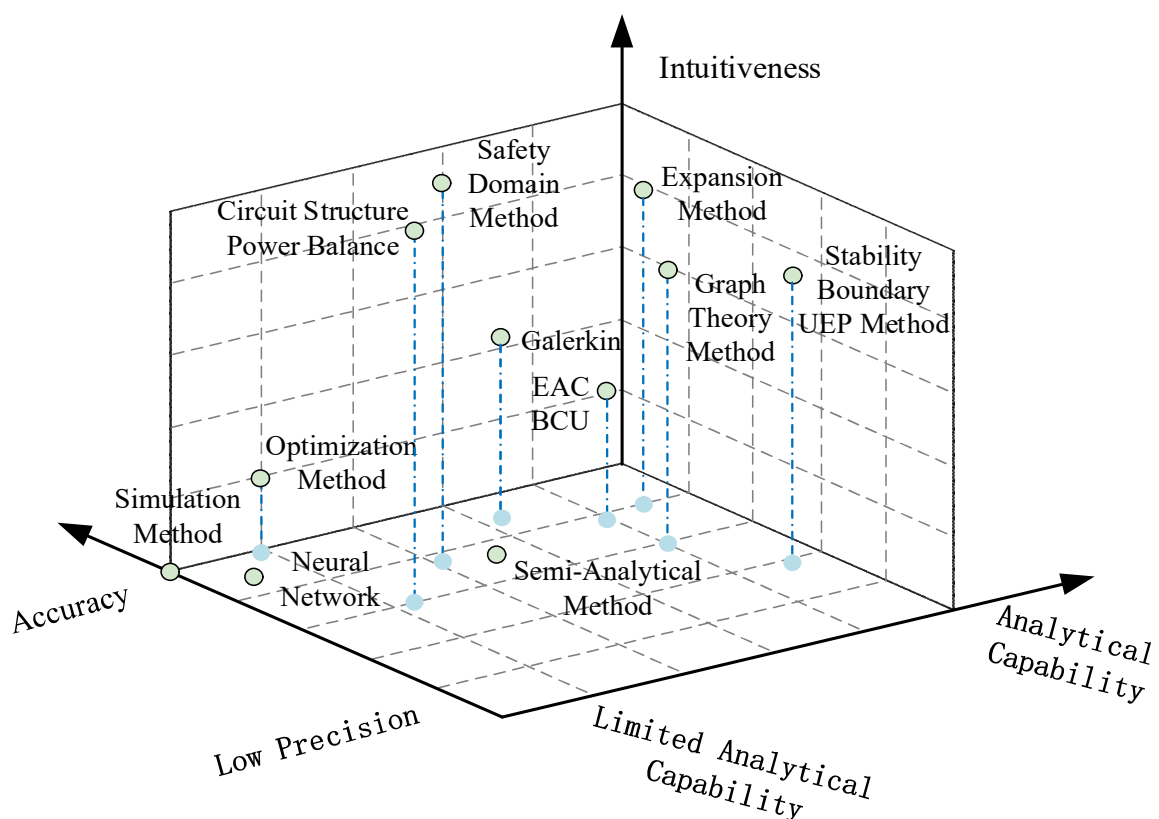


Figure 4. Characteristics of analysis methods for transient stability in power systems.

The integration of renewable energy sources, such as wind and solar power, along with the shift towards hybrid power systems, has introduced new challenges in maintaining power system stability. Renewable energy sources are inherently intermittent and less predictable compared to traditional fossil fuel-based generation. This variability complicates the balancing of supply and demand and can exacerbate transient stability issues.

Additionally, hybrid power systems, which combine alternating current and direct current transmission technologies, further increase the complexity of stability analysis. These systems require advanced control strategies to manage the interactions between different types of generation and transmission technologies, ensuring that the entire system remains stable under various operating conditions.

Furthermore, under the new paradigm of hybrid AC-DC grids with high penetration of wind and solar power, the power system is in a transitional phase. The grid's sending and receiving ends are highly coupled [24], and stability issues related to voltage, frequency, and power angle are intertwined [25]. The presence of multiple large disturbance sources, significant consequences of disturbances, and numerous uncertainties—along with high proportions of renewable energy and power electronic devices [26]—are reshaping the grid's stability characteristics from a more microscopic perspective and shorter time scales, emulating the stability mechanisms of synchronous machines [27]. Thus, the transient stability inherent in synchronous machines should be revisited. The primary goal of this review article is to provide a comprehensive analysis of the various research methods employed for transient stability analysis under large disturbances in power systems. This article categorizes and evaluates these methods, highlighting their fundamental principles, recent advancements, and applicability to modern power grids. By comparing the advantages and disadvantages of each method, this review aims to offer insights that will guide future research and practical applications in the field of power system stability.

The research methods for large disturbance transient stability analysis are categorized into five main types: simulation methods, direct methods, data-driven methods, analytical methods, and other methods. This classification is based on the fundamental principles, approaches, and objectives underlying each method. However, it is important to note that these methods are not mutually exclusive and often share commonalities and overlaps. Simulation methods use detailed mathematical models of power systems and numerical techniques to simulate the dynamic behavior of the system under transient conditions. They provide time-domain solutions by integrating the system's differential equations step-by-step. Rooted in Lyapunov's stability theory, direct methods determine the stability of a power system by analyzing the system's energy functions and stability margins without requiring time-domain simulations. These methods focus on the initial conditions and the potential energy of the system. Leveraging the advancements in machine learning and artificial intelligence, data-driven methods use historical and simulated data to train models that can predict the stability of power systems. These methods do not require detailed system models but instead rely on data patterns and statistical analysis. Analytical methods involve mathematical techniques to derive approximate analytical expressions for the system's response to disturbances. They often use perturbation theories, asymptotic expansions, and other mathematical tools to obtain high-precision approximate solutions. Other methods include hybrid approaches that combine elements from the above methods or introduce new techniques not covered by the other four categories. Examples include methods that integrate optimization techniques with stability analysis or use novel mathematical frameworks to address complex stability problems.

Despite their distinct classifications, there are several commonalities and overlaps among these methods, as follows.

(1) Hybrid Approaches: Many modern stability analysis techniques combine elements from multiple categories. For example, simulation methods can be enhanced with data-driven techniques to improve accuracy and reduce computational burden. Similarly, analytical methods can incorporate data-driven insights to refine their models.

(2) Use of Data: Both data-driven methods and simulation methods rely heavily on data. Simulation methods generate detailed time-series data through numerical simulations, while data-driven methods utilize these data to train predictive models. This shared reliance on data creates opportunities for integrating these approaches.

(3) Mathematical Foundations: Analytical methods and direct methods both leverage mathematical theories to understand system behavior. Direct methods use Lyapunov



functions, while analytical methods use perturbation theory and other mathematical tools. These common mathematical foundations can facilitate the development of hybrid methods that draw on the strengths of both approaches.

(4) Modeling Complexity: Simulation methods and analytical methods both involve creating detailed models of power systems. While simulation methods focus on time-domain solutions, analytical methods seek to simplify these models to obtain tractable solutions. The modeling techniques used in both approaches often overlap, particularly in the initial stages of model development.

(5) Goal of Stability Assessment: All methods ultimately aim to assess the stability of power systems under large disturbances. Whether through time-domain simulations, energy function analysis, machine learning predictions, or mathematical approximations, the end goal is to ensure the reliability and security of power systems. This shared objective drives innovation and integration across different methods.

The remainder of this paper is organized as follows. Section 2 discusses the simulation methods, detailing their evolution and current state. Section 3 covers direct methods, including their theoretical foundations and practical applications. Section 4 explores data-driven methods, focusing on recent advancements in artificial intelligence and machine learning. Section 5 examines analytical methods, including asymptotic expansion and intrusive approximation techniques. Section 6 presents other integrated methods that combine multiple approaches for enhanced stability analysis. Finally, Section 7 concludes the paper with a summary of findings and recommendations for future research directions.

## 2. Simulation Methods

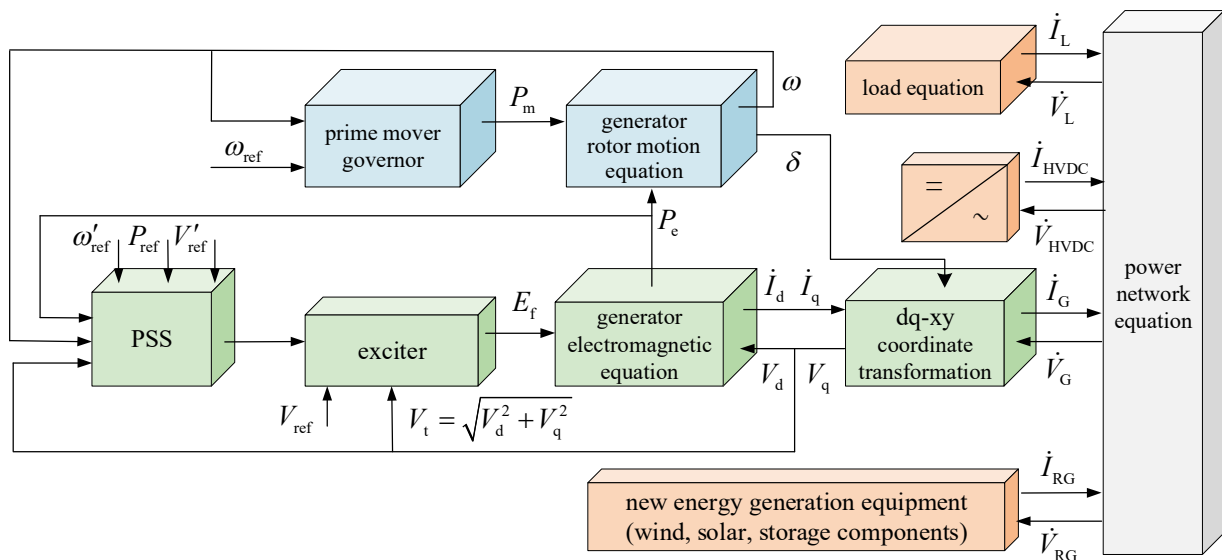
Simulation methods have accompanied the development of power systems throughout their various stages. Early on, analog network analyzers [28] were invented to compute voltages and power flows under normal operating conditions and transient power angle responses under fault conditions. In 1957, the deployment of a large-scale power flow program on the IBM704 digital computer by American Electric Power Service Corporation [29] marked the beginning of large-scale digital computing in power systems. In 1972, Dommel H.W. and Sato N. introduced a numerical integration method based on the implicit trapezoidal rule [30], which established the mathematical foundation for transient stability simulation and is still used today. In essence, the mathematical model of a power system can be represented by a set of differential-algebraic equations as follows:

$$\begin{cases} \dot{x} = f(x, y) \\ 0 = g(x, y) \end{cases} \quad (2)$$

where the state variable  $x$  describes the system's dynamic characteristic, the algebraic variable  $y$  represents the system parameters, and the state function  $f$  and algebraic function  $g$  are defined accordingly. By establishing differential equations for various components of the power system and connecting them through the power network, and using initial value information after major disturbances, the dynamic response of the power system can be obtained through step-by-step integration of Equation (2). Overall, the simulation method for transient stability in power systems primarily encompasses three research dimensions: power system models, numerical algorithms, and simulation software.

In terms of power system models, significant advancements have been made since the 1970s in modeling the four main components: generators, excitation systems, governor control systems, and loads. This progress led the Institute of Electrical and Electronics Engineers (IEEE) to release relevant standard models [31]. In the new century, various component models have continued to evolve. For instance, reference [32] summarizes models and implementation methods for converters, DC switches, DC loads, and DC grid system controls. Reference [33] compares quasi-steady-state models of converters with classical models, defining the applicability of HVDC models for practical engineering. Reference [34] introduces an accurate control model and control strategy for doubly fed wind generators that accounts for dynamic stator excitation currents, making it suitable for harsh

conditions such as sudden voltage drops in the grid. Reference [35] proposes a simulation model for large photovoltaic power plants using an online equivalence method, resulting in the simplest photovoltaic model with acceptable simulation accuracy. Reference [36] presents a storage system model that considers battery charge/discharge power and cycle limits, showing accurate external characteristic trends under external disturbance tests. The development and refinement of these emerging component models, including DC systems, wind turbines, photovoltaic systems, and energy storage, have enriched the simulation logic of modern power grids, as illustrated in Figure 5.



**Figure 5.** Power system simulation logic diagram.

In the realm of power system numerical algorithms, several improved simulation algorithms have been proposed to accelerate simulation speed. Reference [37] discusses a range of advancements in this area. Reference [19] introduced a dynamic multi-dimensional order control method for transient stability calculations, tailored to the varying dynamic characteristics of state variables and computational accuracy constraints during simulations. Reference [38] simplified the iterative calculation of node voltages, proposing an adaptive step-size Newton method that enhances the speed of transient stability simulations. Reference [39] proposed a spatial partitioning algorithm based on hierarchical recursion, which is suitable for parallel computing of transient stability. Reference [40] restructured the network equations to exhibit diagonal partition characteristics and introduced a dynamic parallel simulation algorithm based on preconditioned conjugate gradient methods. These advancements in both serial and parallel numerical algorithms, coupled with rapidly evolving computer hardware [41], have significantly improved the computational speed of transient stability simulations and deepened the application of simulation methods in the industry. In the realm of simulation software, the Bonneville Power Administration (BPA) under the U.S. Department of Energy pioneered the development of large-scale power system offline analysis programs in the 1960s. Over the years, this has led to the creation of numerous representative simulation software globally. Notable international simulation software includes PSS/E V34, DIgSILENT V2024, and BPA V5.8. In China, the Power System Analysis Software Package (PSASP) is among the prominent domestic simulation tools. These industrial-grade software packages are used for studying the transient stability of actual power systems, including voltage, angle, and frequency, as well as for planning and operating large AC-DC interconnected grids. They represent the full potential of simulation methods in practical applications.

However, merely conducting simulation calculations and obtaining curves that reflect the dynamic processes of the power grid is insufficient to meet engineering needs. A



robust stability analysis method must address two fundamental issues: defining stability margin indicators and establishing a clear analytical relationship between grid parameters and stability margins, with this relationship being as simple and physically meaningful as possible. Simulation methods typically provide intuitive time-domain curves of various electrical quantities but struggle to quantitatively explore the analytical relationship between grid stability margins and parameters. Consequently, approximate results are often used, such as the concepts of equal-area criteria, short-circuit capacity, inertia, and power balance principles, to perform approximate transient stability analysis alongside simulation methods. For instance, references [42,43] analyze the impact of “wind-fire bundling” systems on the transient stability of thermal power units using the concepts of equivalent mechanical inertia and equal-area criteria, explaining the mechanisms by which wind turbines and photovoltaic stations worsen transient angle stability in simple network configurations. Reference [44] analyzes the interactive stability mechanisms in wind–solar–thermal bundled systems based on grid connection point short-circuit capacity and explains the impact of wind and photovoltaic power fluctuations on the grid using unbalanced power. These approaches, combining “quantitative simulation with qualitative analysis”, highlight the deep demand for understanding power system stability mechanisms and parameter relationships in the context of simulation and indicate future directions for the development of simulation methods.

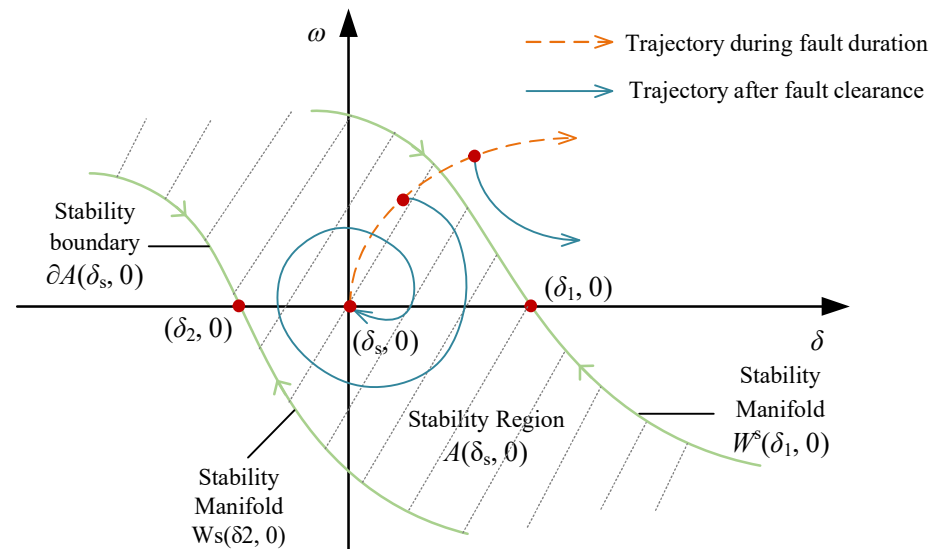
In recent years, as power systems have exhibited characteristics of large-scale ultra-high voltage AC-DC interconnected grids, higher demands have been placed on power system simulation. Modern power system simulation is evolving towards higher dimensions, broader scope, deeper analysis, and greater accuracy [45]. Developing tools with high-resolution relationships, systemic comprehensiveness, high levels of visualization, and robust stability analysis will be the focus of future advancements in power system simulation methods.

### 3. Direct Methods

In the 1950s, when computer resources were extremely limited, overcoming the computational burden of step-by-step integration methods for power systems required innovative approaches. More than 20 years after the popularization of the equal-area criterion (EAC) [46], Magnusson and others extended the concept of transient energy from single-machine systems to three-machine systems [47]. They followed the EAC paradigm to perform transient stability analysis, allowing the direct determination of the system’s transient stability by manually calculating the initial rotor values during faults. Over time, it became clear that the concept of transient energy in the EAC was merely a special case of applying Lyapunov’s theory to single-machine systems. With the publication of related work by Dr. Gless and El-Abiad in 1966 [48,49], direct methods based on Lyapunov’s theory were formally introduced into power system transient stability analysis. The appeal of direct methods lay in their “foresight” characteristic, focusing only on the initial conditions of faults, as illustrated in Figure 6. More importantly, as the power grid evolved into larger-scale configurations, particularly after the massive blackout in the Northeastern United States in 1965 [50], there was increasing demand from the industry for online security and stability analysis of power systems [51]. This led to a surge in research enthusiasm for direct methods, focusing on aspects such as the analytical construction of energy functions [19] and the calculation of critical energy [52].

In the area of analytical construction of energy functions, the first integral method [53] can be used to construct energy functions for simplified network models. However, the potential energy function constructed by this method depends on the path of the integral term, and the positive definiteness of the energy function cannot be strictly proved. Reference [54] obtained an energy function that accounts for generator excitation dynamics using the method of energy conservation, although its applicability to large systems still requires further verification. Reference [55] used the variable gradient method to construct an energy function, highlighting the provision of a systematic construction process but

facing difficulties in determining the coefficients of related functions. Reference [56] constructed the system's energy function using ordinary differential methods, transforming the problem of constructing the energy function into an optimization problem of nonlinear overdetermined equations; thus, the existence and uniqueness of the solutions to these ordinary differential equations need further exploration. These various forms of energy functions either lack precision or raise questions about their reliability. It has gradually been recognized that analytical energy functions for complex power system models are nearly non-existent, leading the development of energy function construction towards model simplification or numerical approaches.



**Figure 6.** Diagram of the direct method.

In terms of determining critical energy, there are primarily two approaches. One is the potential energy boundary surface (PEBS) method [57], which aims to overcome the difficulties in solving for unstable equilibrium points (UEPs). This method uses the maximum potential energy on the trajectory of a persistent fault as the critical energy, approximating the stability boundary with an equal energy surface. Practical experience shows that the PEBS method can sometimes yield overly optimistic results in certain fault scenarios. Another approach is the boundary of stability region-based controlling unstable equilibrium point (BCU) method [58], which defines the system's dominant unstable equilibrium point (CUEP) as the escape point on the stability boundary for the trajectory of persistent faults. The energy value at the CUEP is used as the critical energy, with the equal energy surface approximating the stability boundary. The BCU method partly overcomes the conservatism of the PEBS method and has been further developed for complex power system models, including those incorporating excitation systems [59].

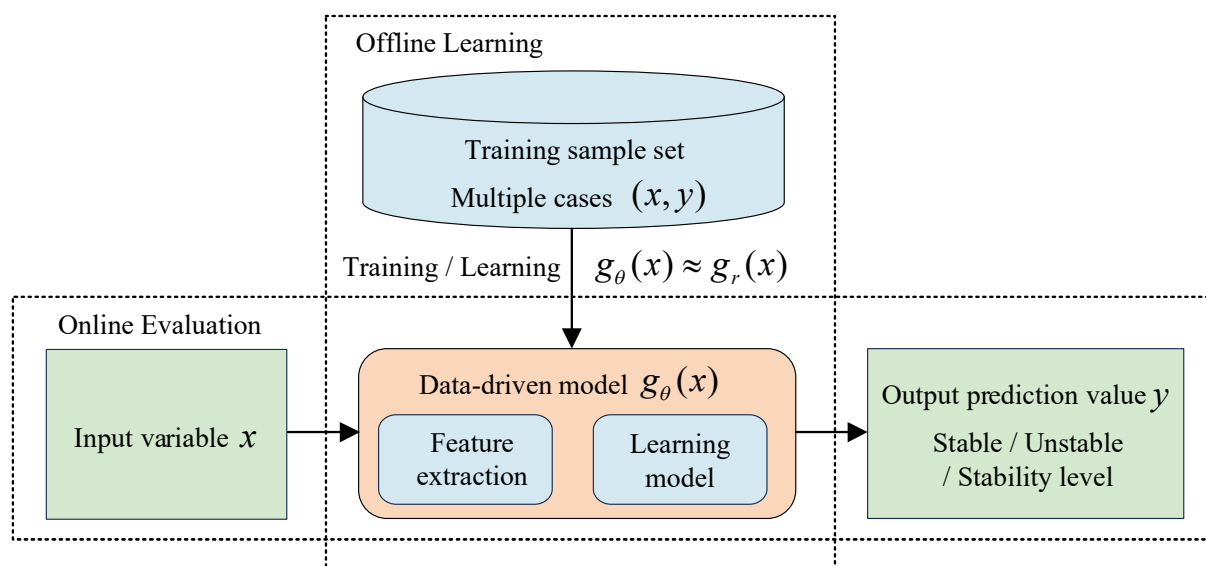
Once the expression for the energy function  $V$  and the critical energy function value  $V_{cr}$  are obtained, let  $\omega_c$  and  $\delta_c$  represent the rotor angular velocity vector and rotor angle vector at the moment of fault clearance, respectively. If the system's transient energy  $V(\omega_c, \delta_c)$  at the fault clearance moment does not exceed the critical energy function value  $V_{cr}$ , i.e.,  $V(\omega_c, \delta_c) \leq V_{cr}$ , then the system is considered stable. Conversely, if  $V(\omega_c, \delta_c) > V_{cr}$ , the system is deemed unstable. Furthermore, the energy function difference  $\Delta V = V_{cr} - V(\omega_c, \delta_c)$  can be interpreted as the stability margin or energy margin. Note that  $\omega_c$  and  $\delta_c$  are initial values  $\mathbf{X}_0$  in Equation (1) and can be seen as functions of the control variable  $\mathbf{u}$ . Therefore, parameter stability analysis can be conducted based on sensitivity information  $\partial \Delta V / \partial \mathbf{u}$ .

Overall, the direct method, based on Lyapunov theory, has a clear physical concept and solid theoretical foundation, but it also faces evident difficulties, such as the construction of the energy function and determination of critical energy. Additionally, various algorithms

developed to address these issues also introduce a certain computational burden. Future advancements in the practical application of the direct method will aim towards more convenient computations and more refined theories.

#### 4. Data-Driven Methods

Since Dr. Dy-Liacco introduced pattern classification as a representative data-driven method for power system security and stability analysis in 1968 [60], the standard application paradigm has remained largely unchanged. Particularly with the widespread deployment of phasor measurement units (PMUs) using Global Positioning System (GPS) for synchronized timing, real-time acquisition of power system data has become more feasible, leading to increased attention being paid to data-driven methods in recent years [61]. As illustrated in Figure 7, assuming that there is a real correspondence between the power system operational data  $x$  and the stability state  $y$ , represented by a mapping  $y = g_r(x)$ , the core of data-driven methods is to mine the mapping relationship (6) between the system operational state  $x$  and the stability condition  $y$  from offline data. This mapping  $g_\theta(x)$  aims to approximate  $g_r(x)$  as closely as possible. With this mapping, stability evaluation can be performed based on the current system information input value  $g_\theta$  in an online setting. Generally, data-driven methods follow a standard process including data preparation, feature selection, model training, and online evaluation stages [62–64]. The primary research dimensions in data-driven methods are data and algorithms.



**Figure 7.** Application paradigm of data-driven methods in power system stability.

The first research dimension is data generation, which forms the foundation for data-driven methods. Despite the vast amount of operational data accumulated from decades of real power system operations, data suitable for transient stability analysis are scarce. This scarcity is due to the extreme lack of samples under large disturbances leading to instability in actual power systems. As a result, simulated data must be generated through time-consuming time-domain simulations. Simulation-based methods require predefined base load flows and fault information. Given the vast search space in large-scale power system scenarios, issues like the “curse of dimensionality” and “combinatorial explosion” arise. Reference [65] utilized Monte Carlo random sampling methods to generate operating points, simplifying the search space and enhancing sample generation efficiency. Reference [66] employed vine copulas methods to generate more representative system state data, which aid in training better sample generation models. Reference [67] used importance sampling strategies to maximize sample generation while reducing computational burdens. Reference [68] applied convex optimization methods to exclude operating

points far from the stability boundary, producing sample data that better represent the characteristics near the stability boundary. These methods have alleviated the pressure of generating vast amounts of samples to some extent, but efficiently and quickly generating more representative data that reflect power grid stability characteristics remains a challenge. Additionally, the lack of reliable transient stability margin indicators makes it challenging to label large volumes of transient stability data samples.

The second research dimension is the evaluation of intelligent algorithms, which has become a focal point in transient stability analysis. Especially since the advent of deep learning, which has spurred a new wave of artificial intelligence research [69], emerging intelligent evaluation algorithms have been continuously proposed alongside traditional algorithms such as artificial neural networks [70] and decision trees [71]. For instance, reference [72] introduced a transient stability assessment algorithm based on a radial basis function kernel support vector machine (SVM) classifier, which reduces feature dimensionality while ensuring effective stability assessment. Reference [61] proposed a stability discriminator with strong noise resistance using a stacked variational autoencoder (VA) method, enhancing the generalization ability of transient stability models. Reference [73] employed multi-scale convolutional kernels to directly use low-level measurement data as input features, constructing an end-to-end convolutional neural network (CNN) transient stability evaluator. These intelligent evaluation algorithms, represented by SVM, VA, and CNN, have either improved stability assessment accuracy, accelerated model training speed, or enhanced model robustness. However, they have not addressed two fundamental issues. First, the stability assessment mechanisms of these black box data models lack interpretability. Traditionally, mechanistic models have guided power grid operations with their intuitive and reliable characteristics, leading to skepticism about data-driven methods with unclear mechanisms. Second, the prediction accuracy of these data models still needs improvement. Although various developed intelligent evaluation methods perform well on existing datasets, their ability to maintain this performance in the diverse and variable scenarios of actual power grid operations remains a concern.

In recent years, non-intrusive approximation methods have garnered considerable attention. As indicated by Equation (1), power system stability analysis can be understood as solving a parametric equation. Non-intrusive approximation methods can directly focus on parameter solutions  $X(t; u)$  and  $y(t; u)$ . The basic idea is to presuppose the structure of the system's parameter solution and then determine the coefficients using specific algorithms and data to obtain an approximate expression of the system's parameter solution. This method does not require establishing a system model; it only needs simulation data at certain parameter points.

For instance, reference [74] performed rational function fitting between the photovoltaic output voltage and inverter control parameters during fault and recovery phases, resulting in a rational function analytical expression for transient voltage. Reference [75] assumed that the voltage trajectory consists of parametric polynomial functions and used collocation methods to obtain the analytical expression of the polynomial trajectory, which can then delineate the power grid stability boundary. Reference [76] employed the least squares method to derive a polynomial expression for the rotor motion of a power system and used it to analyze the dominant oscillation modes in a multi-machine system. These methods are straightforward and clear, allowing direct acquisition of parameter solutions for power grid responses based on simulation data and approximation techniques, thus facilitating stability analysis work. They hold significant research value but also face the "black box" issue, lacking interpretability and theoretical understanding. For example, when fitting voltage dynamics, it remains unclear whether polynomial structures are superior to rational function structures, and whether parameter solutions derived from different structures are consistently valid across the entire interval. These fundamental questions still require further investigation.

Overall, data-driven methods for power system stability analysis have been a significant focus in the industry over the past decade. However, this topic still faces two main

challenges. First, there are numerous intelligent evaluation algorithms and non-intrusive approximation methods, and it remains unclear which algorithms are suitable for power systems while achieving satisfactory and convincing performance, including interpretability [77] and predictive accuracy. Establishing a mathematical foundation at the theoretical level is necessary. Second, there is the challenge of simplifying obtained parametric expressions to extract qualitative or quantitative insights into power system stability. This involves deriving knowledge from data [78] to advance the progress of stability analysis.

## 5. Analytical Methods

In this section, the roles of various analytical methods in transient stability analysis are specifically elucidated. Section 5.1 analyzes the asymptotic expansion method and presents comparison curves. Section 5.2 discusses the intrusive approximation method and its fundamental principles. Section 5.3 describes other types of analytical methods.

### 5.1. Asymptotic Expansion Method

The method of asymptotic expansion is a widely used technique in mathematics for obtaining high-precision approximate solutions to equations or systems of equations. As early as the 19th century, Henri Poincaré and others developed a set of asymptotic expansion methods based on small parameters while studying complex celestial orbits [79]. The core idea of this method is to express the solution curve as a finite series of terms in a small parameter. Consider solving the following initial value problem for a differential equation:

$$\begin{cases} \dot{x} = f(x, y) \\ 0 = g(x, y) \end{cases}, \quad (3)$$

where the right-hand side term represents a mapping involving small parameter  $\epsilon$  and  $x(t)$ . The asymptotic expansion method assumes that the approximate solution curve is composed of a superposition of a set of asymptotic curves, as shown below:

$$\tilde{x}(t) = \sum_{i=0}^n \epsilon^i \hat{x}_i(t), \quad (4)$$

where  $\epsilon$  represents the small parameter, and  $\epsilon \ll 1$ ;  $\tilde{x}(t)$  represents the perturbed solution curve;  $\hat{x}_i(t)$  represents the  $i$ -th curve; and  $n$  is the total number of curves. By substituting Equation (4) into Equation (3) and organizing according to the order of the small parameter  $\epsilon$ , a series of equations can be derived to solve for the curve  $\hat{x}_i(t)$ . After superimposing, the perturbed solution  $\tilde{x}(t)$  can be obtained.

Generally, the asymptotic expansion method requires the nonlinear terms in the equations to exhibit characteristics of small parameters and is primarily applicable to systems with relatively low degrees of nonlinearity. Furthermore, the approximate solution curve given by Equation (4) often contains secular terms (SC), which can cause the expanded solution to fail to capture the dynamic response of the actual system. Specifically, consider the following initial value problem [80]:

$$\begin{aligned} x'' + \epsilon x' + x &= 0; \\ x(0) &= 0, \quad x'(0) = 1, \end{aligned} \quad (5)$$

where  $\epsilon > 0$  represents the small parameter. Substituting Equation (4) into Equation (5) yields the expanded solution as follows:

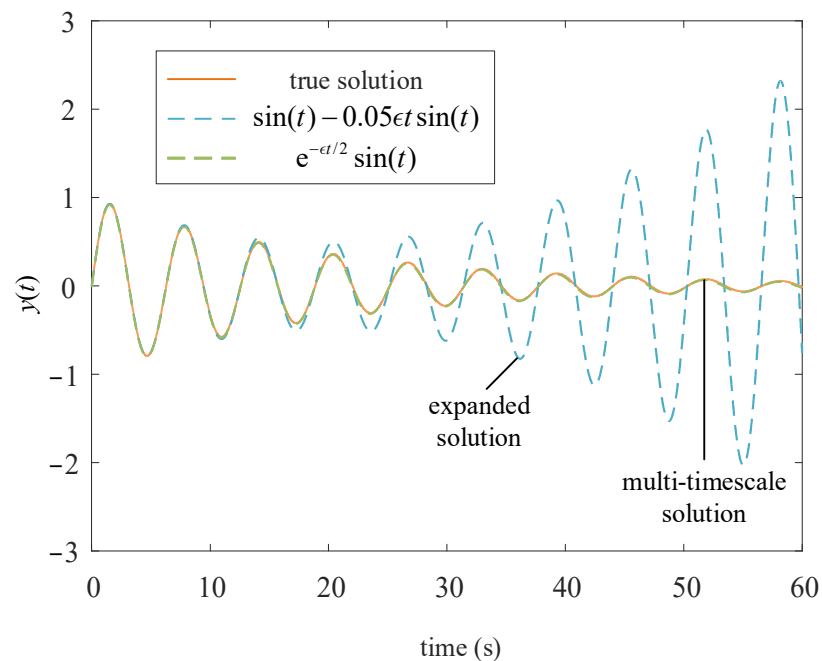
$$\tilde{x}(t) = \hat{x}_0 + \epsilon \hat{x}_1(t) + \mathcal{O}(\epsilon) = \sin(t) - \epsilon \frac{1}{2} t \sin(t) + \mathcal{O}(\epsilon), \quad (6)$$

where the symbol  $\mathcal{O}(\epsilon)$  denotes higher-order infinitesimals of the small parameter  $\epsilon$ . It is noted that  $\hat{x}_1(t)$  contains  $t \sin(t)$ , which is called the secular term. As time approaches

infinity, this term will cause oscillatory divergence. In fact, the exact analytical solution to Equation (5) is as follows:

$$x(t) = \frac{1}{\sqrt{1 - \epsilon^2/4}} e^{-\frac{1}{2}\epsilon t} \sin(t\sqrt{1 - \frac{\epsilon^2}{4}}). \quad (7)$$

According to Equation (7), the true solution does not exhibit divergent oscillations, indicating that the expanded solution in Equation (6) is inadequate for capturing the true dynamics of the system. The dynamic comparison between the true solution (Equation (7)) and the expanded solution (Equation (6)) is illustrated in Figure 8.



**Figure 8.** Comparison of true solution, expanded solution, and multi-timescale solution.

As shown in Figure 8, the presence of secular terms causes the expanded solution to fail to capture the true dynamic response of the system and may even lead to conclusions that contradict the stability direction of the original system. To address the issue of secular terms encountered in asymptotic expansion methods, various perturbation methods have been developed from different perspectives. The multi-timescale method [81,82], for example, introduces different time scales to characterize the slowly varying features of the system based on the slow variations in amplitude and phase. This method transforms the original nonlinear equation with small parameters into a series of linear ordinary differential equations and avoids the secular term problem in asymptotic methods by solving these ordinary differential equations, thus providing a consistent and effective solution for the system. The averaging method [83,84], based on the idea of constant variation, assumes that the solution of a nonlinear system is similar in form to that of its derived system, considering the quasi-harmonic nature in weakly nonlinear systems. It also assumes that the amplitude and initial phase of the nonlinear oscillatory system are functions of the small parameter, and replaces these with the average value over one period to form averaged equations for amplitude and phase. The renormalization group method [85,86] specifically addresses weakly nonlinear polynomial systems. It separates the secular term directly from the asymptotic solution and transfers the conditions for generating secular terms into initial value conditions through a form transformation, resulting in renormalization equations. Solving these renormalization equations yields a consistent and effective asymptotic solution free from secular terms. Taking the initial value



problem described by Equation (5) as an example, the multi-timescale solution obtained by eliminating the secular terms using the multi-timescale method is as follows:

$$\tilde{x}_0(t_1, t_2) = e^{-\frac{t_2}{2}} \sin(t_1) = e^{-\frac{\epsilon t}{2}} \sin(t), \quad (8)$$

where  $t_2 = \epsilon t$  represents the slow time scale used to characterize slowly varying features.

The comparison between the multi-timescale solution and the true system solution is illustrated in Figure 8. It can be seen that the asymptotic expansion method, after overcoming the secular term problem, can effectively capture the dynamic response and provide an analytical expression. As a result, the application of this method in power systems has also gained increasing interest in recent years. Reference [87] reports early research results of this method in power systems. Reference [88] employed a similar Adomian decomposition method to obtain a semi-analytical solution for differential-algebraic equations and validated the accuracy of this solution in power systems with tens of thousands of nodes. Reference [89] derived series solutions for the dynamic response of power systems, improved the computational efficiency of these series solutions using Padé approximations combined with several vectorization techniques, and validated them in large-scale real power systems. Reference [90] used the differential transformation method to convert nonlinear terms into a series of analytically solvable functions, obtaining corresponding series solutions. A comparison with Adomian decomposition indicates that the differential transformation method enhances the analyticity of grid equations and reduces the computational burden in numerical simulations. Overall, the above methods fit the transient response of power systems using finite series sums, and the obtained series solutions can effectively reflect the dynamics of the original system, enhancing the level of analytical analysis. Another special method within asymptotic expansion is the singular perturbation method. This method utilizes the multi-time scale characteristics present in power systems to perform asymptotic expansion calculations, decomposing the system into fast and slow subsystems. This approach allows for model reduction of the power system on different time scales [91], simplifying system dynamics and highlighting the main behaviors of the system [92].

### 5.2. Intrusive Approximation Method

Unlike non-intrusive approximation methods, such as data-driven approaches which directly train a mapping relationship between power grid stability and input information without focusing on the structure of the original equations, intrusive approximation methods take into account the characteristics of the original dynamic equations. Among these, the Galerkin method [93,94] is one of the most well-known intrusive methods. The Galerkin method is intrusive because it incorporates the structural information of the equations. Its advantages include high accuracy and approximation error that decays with the polynomial degree. In mathematical terms, stability analysis of differential-algebraic Equation (2) can be described in a unified general form as follows:

$$A(u; p) = 0, \quad (9)$$

where  $u = \{u_i\}_i^M$  represents the differential variable  $x$  and the algebraic variable  $y$  in the original differential-algebraic Equation (2);  $M$  denotes the dimension of the original model;  $p = \{p_i\}_i^{N_p}$  denotes the system parameters, with  $N_p$  representing the number of parameters. Choosing a polynomial basis trial basis  $\{\phi_i(p)_{i=1}^N\}$ , the approximated variable is denoted as  $u_j^*(p) = \sum_{i=1}^N c_i \phi_i(p)$ , with  $M \times N$  being the number of undetermined coefficients. Substituting the approximated variable into Equation (9) yields a non-zero residual as follows:

$$R = A(u^*(p); p) = A\left(\sum_{i=1}^N c_i \phi_i(p); p\right), \quad (10)$$

where  $R$  includes both the undetermined coefficient  $c_i$  and system parameter  $p$ , with the dimension of  $R$  being the same as that of the original model, which is  $M$ . To ensure that the residual  $R$  is orthogonal to the test basis  $\{\Gamma_i(p)_{i=1}^N\}$ , the Galerkin Equation (11) must be satisfied as follows:

$$\langle R, \Gamma_k \rangle = 0; \quad k = 1, \dots, N, \quad (11)$$

where  $\{\Gamma_i(p)_{i=1}^N\}$  is called the test basis, which can be constructed appropriately according to the requirements.

According to Equation (11), each residual is subjected to an inner product operation with all basis functions. The inner product operation is a definite integral. When there are multiple parameters  $p$ , the definite integral operation becomes a multiple integral, with the limits of integration depending on the parameters themselves. The dimension of the equation is  $M \times N$ , which matches the number of undetermined coefficients. After performing the inner product operation, the Galerkin equation will no longer contain parameters  $p$ , but only  $M \times N$  undetermined coefficients. Solving Galerkin equations yields the values  $c_i$  of the undetermined coefficients. Substituting these values into the approximation expression  $u_j^*(p) = \sum_{i=1}^N c_i \phi_i(p)$  allows for the analytical solution to be obtained.

Intrusive approximation methods consider the structure of system model and leverage prior experience with system responses to generally provide sufficiently accurate approximate analytical solutions. These methods have gained increasing attention in recent years. Reference [95] used the Galerkin method to obtain polynomial-type expressions for the dynamic trajectories of power systems, offering new insights into quantitatively characterizing the effects of parameter perturbations on dynamic trajectories and performance. Reference [96] observed that synchronous generators exhibit low-pass filtering characteristics over a large motion range, reflecting the quasi-sinusoidal nature of power angle oscillations. Based on a proposed quadratic harmonic balance method, this study obtained intrusive harmonic solutions for the power angle. Reference [97] introduced a polynomial approximation method based on arbitrary sparse bases and generalized Smolyak sparse grid products for studying parameterized problems in power systems, which improved the accuracy of approximating power grid output curves. Reference [98], inspired by the Tricomi curve of the simple pendulum system, used the harmonic balance method to derive an analytical expression for the bifurcation curve of a single-machine system's swing equation, clarifying the stability patterns within the parameter space of infinite single-machine systems.

### 5.3. Other Analytical Methods

Studying stability from the perspective of stability boundaries is another approach. A profound result regarding stability boundaries is given by the following: let  $S$  represent the state space stability domain,  $\partial S$  denote the boundary of the stability domain, and  $W_i^S$  represent the stable manifold of the  $i$ -th unstable equilibrium point. The following relation holds [99]:

$$\partial S \subseteq \cup_i W_i^S, \quad (12)$$

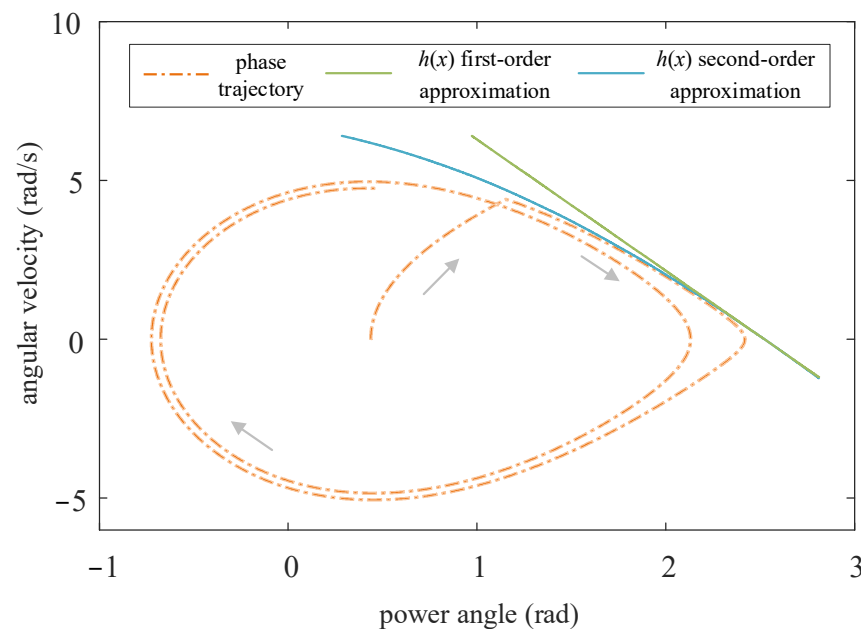
It is evident that the stability boundary  $\partial S$  is contained within the union of the manifolds  $\cup_i W_i^S$ . For more results on the characterization of stability boundaries, refer to reference [100]. Under stronger assumptions, this inclusion can be converted into an equality  $\partial S = \cup_i W_i$ . Unfortunately, stability boundaries are often high-dimensional manifolds, making the computation of unstable equilibrium point (UEP) manifolds still very challenging. This has led researchers to develop various approximation methods for stability boundaries. As early as the 1990s, Fouad A. A. and colleagues began to explore the use of normal form methods to analytically determine the stable manifolds of UEPs in nonlinear power system equations [101–103], which received widespread attention. In the early 2000s, Daizhan Cheng and others developed a theorem related to the stability boundary of Type I UEPs in nonlinear dynamical systems [104], as stated in Theorem 1.

**Theorem 1.** Let  $x_e^1$  represent a Type I unstable equilibrium point of a smooth nonlinear system  $\dot{x} = f(x)$ , and  $\mu$  be the unique positive eigenvalue of the corresponding Jacobian matrix. Then, the stable manifold  $h(x)$  can be locally expressed as:

$$W^S(x_e^1) = \left\{ x \mid h(x) = 0, f^T \frac{\partial h}{\partial x} = \mu h, h(x_e^1) = 0 \right\}, \quad (13)$$

where the rank of the matrix  $\partial h / \partial x$  is 1.

Applying Theorem 1 to the single-machine infinite bus system under typical power grid parameters, the fitting effects of the linear and quadratic approximation parts of Equation (13) are shown in Figure 9. It can be seen that this formula closely approximates the actual boundary. Mei Shengwei and others applied Theorem 1 to power system stability analysis, conducting a series of studies on approximations of the stable manifold boundary [105,106] and transient stability analysis [107,108], which has enhanced the level of analytical theory for power system stability boundaries [109].



**Figure 9.** Diagram of the approximate formula for the stability boundary of a single-machine system.

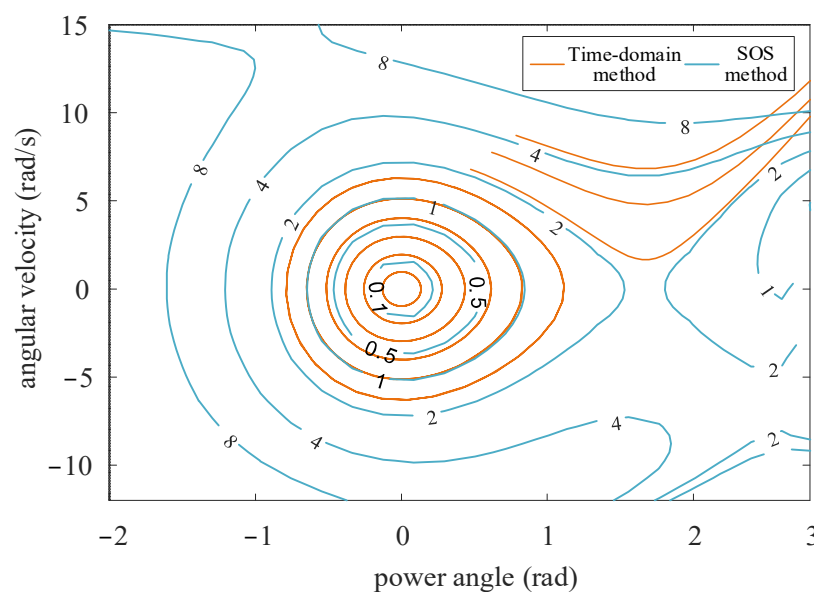
Notably, analytical methods employ mathematical techniques to derive approximate analytical expressions for a system's response to disturbances, achieving a certain level of precision, but also presenting various limitations and assumptions. Perturbation theories assume that a system's behavior can be expressed as a series expansion around a small parameter, which restricts their applicability to nonlinear systems. Additionally, many analytical methods assume linearity or use simplified linear models, which may not accurately capture the complex dynamics of real-world power systems, making their solutions valid only under specific conditions. These methods often rely on idealized conditions and simplifications, such as neglecting higher-order interactions or assuming constant system parameters, leading to discrepancies between analytical results and actual system behavior. While analytical methods aim for closed-form solutions, the derivation can be mathematically intensive and computationally complex, limiting their practicality for real-time applications and large-scale power systems.

In addition, utilizing the inherent characteristics of power grids for stability analysis is a relatively new approach, with analytical graph theory being one such method that has garnered increasing attention in recent years. Dörfler et al. applied the Kron reduction theorem to the context of power networks [110], and established mathematical connections

between graph theory, algebra, spectral distribution, and sensitivity from the perspective of analytical graph theory. Reference [111] used the graph-theoretical properties of power networks, specifically the second smallest eigenvalue of the Laplacian matrix, to establish sufficient conditions for transient stability from an algebraic perspective. Reference [112] improved the synchronization conditions proposed in [111] by incorporating network parameters and topological information, further reducing conservativeness. It was found that network stability can be maintained when the coupling strength of the network exceeds the asynchrony effect. In the context of power systems, this means that if the network is sufficiently tightly connected and the output of synchronous machines is low enough, transient stability issues will not arise [113]. These novel and interesting perspectives provide new insights into stability analysis.

## 6. Other Methods

While various methods have developed independently, their integration often leads to the emergence of new approaches. For example, reference [114] combines simulation and direct methods. By utilizing the kinetic and potential energy functions defined by the direct method and leveraging simulation data, this approach proposes a trajectory analysis method that does not depend on critical energy calculations. The combination of optimization and simulation methods has led to the development of the stability-constrained optimal power flow (SOPF) problem [115], which has attracted considerable research interest. The integration of analytical and simulation methods has resulted in a semi-analytical method suitable for large-scale power system simulations [116], which also serves as a validation tool for time-domain simulation methods. Furthermore, the fusion of optimization and direct methods has given rise to a numerical approach based on sum of squares (SOS) programming [117]. This method, rooted in algebraic geometry, focuses on polynomial dynamic systems and systematically addresses Hilbert's 17th problem [118] through numerical optimization, providing a theoretical foundation for constructing Lyapunov functions. Applying the SOS method to a single-machine system under typical power grid parameters results in contour plots of the region of attraction, as shown in Figure 10. These plots illustrate that the SOS method can effectively delineate the region of attraction in a single-machine system.



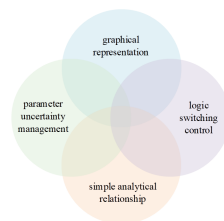
**Figure 10.** The attraction domain of the SOS method in a single-machine system.

Overall, the combination of various methods can achieve complementary advantages. For example, trajectory analysis methods rely on numerical simulations to avoid the calculation of the critical energy function required by direct methods. The SOPF combines

optimization methods to avoid simulating each operating point of the power grid to determine the optimal state. However, these hybrid methods have not overcome their inherent limitations. For instance, methods combining optimization techniques, such as SOS and SOPF, although inheriting the strong systematic advantages of optimization methods, still suffer from the low analytical degree inherent to optimization approaches. This makes it difficult to directly conduct stability analysis, thus positioning these methods as supplementary to numerical computation such as simulation methods in power grid analysis. Despite these limitations, the integration and regeneration of various methods, achieving complementary strengths, remains a fascinating and promising research direction in power system stability analysis.

Recent advancements in transient stability analysis have been propelled by high-performance computing and advanced simulation tools, enabling more accurate power system dynamics simulations. Innovative methodologies, such as hybrid simulation-analytical approaches and the power system digital twin (PSDT), have significantly improved stability assessments and operator training [119]. The integration of artificial intelligence and machine learning has further revolutionized the field, allowing for real-time predictive models and optimized grid stabilizing devices. However, challenges remain, including the need for robust methods to handle renewable energy variability, the impact of cyber-physical interactions, and the development of models for hybrid AC-DC grids. Additionally, standardized protocols and tools are essential for enhancing the reliability and comparability of transient stability analysis methods.

Moreover, in the context of hybrid power grids, the logical switching characteristics of converters [120,121] and parameter uncertainties [122] further exacerbate the difficulty of transient stability analysis. The ideal transient stability analysis technology for hybrid power grids is still under development and should possess several key features, summarized in Figure 11 below.



**Figure 11.** Key features of an ideal transient stability analysis technique.

## 7. Conclusions

In this paper, a comprehensive review and systematic analysis framework for transient stability analysis of power systems under large disturbances are established from various methodological perspectives. The main methods examined include simulation methods, direct methods, data-driven methods, analytical methods, and other integrated methods. The conclusions are as follows:

(1) **Simulation Methods:** The evolution of simulation methods is traced from analog network analyzers to sophisticated modern simulation software. Significant advancements in power system models, numerical algorithms, and simulation software have significantly enhanced the accuracy and applicability of these methods in practical power system stability analysis.

(2) **Direct Methods:** Rooted in Lyapunov's theory, direct methods offer a solid theoretical foundation for transient stability analysis. Despite challenges in constructing energy functions and determining critical energy, advancements such as the potential energy boundary surface and boundary of stability region-based controlling unstable equilibrium point methods have improved the practical applicability of direct methods.

(3) **Data-Driven Methods:** Leveraging advancements in artificial intelligence and machine learning, data-driven methods have become increasingly prominent. These methods focus on mining the mapping relationships between system operational states and stability

conditions using vast amounts of operational and simulated data. However, challenges remain in generating representative data and improving the interpretability and predictive accuracy of intelligent evaluation algorithms.

(4) Analytical Methods: Methods like the asymptotic expansion, intrusive approximation, and stability boundary methods offer valuable insights into the dynamics of power systems. Techniques such as the multi-timescale method, averaging method, and Galerkin method have been particularly effective in capturing the dynamic responses and stability characteristics of power systems.

(5) Other Methods: The integration of various approaches often leads to innovative solutions. For example, the combination of simulation and direct methods has resulted in trajectory analysis methods that do not depend on critical energy calculations. The stability constrained optimal power flow problem integrates optimization and simulation methods, avoiding the need to simulate each operating point of the power grid. Additionally, the fusion of optimization and direct methods has given rise to numerical approaches based on sum of squares programming, providing a theoretical foundation for constructing Lyapunov functions and delineating the region of attraction in power systems.

Overall, this paper provides a mechanistic understanding of power system transient stability under large disturbances and offers a detailed comparison of different research methods, as shown in Table 1. Future research will focus on integrating these methods to develop more robust and accurate stability analysis techniques, considering the dynamic evolution of power systems and the increasing complexity caused by high penetration of renewable energy sources.

**Table 1.** Summary and comparison of different transient stability analysis methods.

| Methods             | Principles                                                                                                       | Advantages                                                                                                                                                                                                                         | Disadvantages                                                                                                                                                                                                                                   | Future Directions                                                                                                                                               |
|---------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Simulation Methods  | Use step-by-step integration to directly obtain numerical solutions and capture the system's nonlinear dynamics. | <ul style="list-style-type: none"> <li>• Provide intuitive time-domain curves of various electrical quantities.</li> <li>• Well-developed models and numerical algorithms enhance accuracy and applicability.</li> </ul>           | <ul style="list-style-type: none"> <li>• High computational burden, especially for large-scale systems.</li> <li>• Difficulty in quantitatively exploring the analytical relationship between grid stability margins and parameters.</li> </ul> | Develop tools with high-resolution relationships, systemic comprehensiveness, high levels of visualization, and robust stability analysis.                      |
| Direct Methods      | Based on Lyapunov's theory, it focuses on the initial conditions of faults to determine transient stability.     | <ul style="list-style-type: none"> <li>• Clear physical concept and solid theoretical foundation.</li> <li>• Can provide stability margins and energy margins directly.</li> </ul>                                                 | <ul style="list-style-type: none"> <li>• Difficulties in constructing the energy function and determining critical energy.</li> <li>• Computational burden due to complex algorithms.</li> </ul>                                                | More convenient computations and more refined theories to enhance practical applicability.                                                                      |
| Data-Driven Methods | Mine the mapping relationship between system operational state and stability condition from offline data.        | <ul style="list-style-type: none"> <li>• Leverage vast amounts of operational and simulated data for stability analysis.</li> <li>• Can handle large datasets and complex patterns through machine learning algorithms.</li> </ul> | <ul style="list-style-type: none"> <li>• Scarcity of representative data for large disturbances.</li> <li>• Lack of interpretability and predictive accuracy in intelligent evaluation algorithms.</li> </ul>                                   | Improve data generation techniques and labeling accuracy and enhance interpretability and predictive performance of intelligent algorithms.                     |
| Analytical Methods  | Use mathematical techniques to derive approximate analytical expressions for system responses and stability.     | <ul style="list-style-type: none"> <li>• Provide high-precision approximate solutions and analytical expressions.</li> <li>• Can effectively capture the dynamic responses and stability characteristics.</li> </ul>               | <ul style="list-style-type: none"> <li>• Requires the nonlinear terms to exhibit characteristics of small parameters.</li> <li>• May encounter issues with secular terms causing oscillatory divergence.</li> </ul>                             | Develop more robust techniques to handle highly nonlinear systems and combine analytical methods with other approaches for enhanced accuracy and applicability. |



Table 1. Cont.

| Methods       | Principles                                                                                   | Advantages                                                                                                                                                                                                                                 | Disadvantages                                                                                                                                                                 | Future Directions                                                                                                                                                                                            |
|---------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Other Methods | Combine various approaches to leverage their complementary strengths for stability analysis. | <ul style="list-style-type: none"> <li>Can achieve complementary advantages by integrating different methods.</li> <li>Offer innovative solutions like trajectory analysis and stability-constrained optimal power flow (SOPF).</li> </ul> | <ul style="list-style-type: none"> <li>Inherits limitations from the combined methods.</li> <li>Low analytical degree and difficulty in direct stability analysis.</li> </ul> | Further explore the integration of different methods for comprehensive stability analysis and develop hybrid approaches tailored for modern power systems with high penetration of renewable energy sources. |

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